

Existence of Global Solutions for the Interpolated Stochastic Heat and Wave Equation with Super-Linear Coefficient

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Special Session on
AU Stochastic Seminar

March 14, 2022

$$\left\{ \begin{array}{ll} \left(\partial_t^\beta + \frac{\nu}{2} (-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & \beta \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & \beta \in (1, 2). \end{array} \right. \quad (0.1)$$

References